

Muhammad Hamza

Home: 31200, Layyah, Pakistan

Email address: hamzamuhammad.research@gmail.com

Website: https://www.github.com/zahqresh **WeChat:** mhamzaqureshi

Gender: Male **Date of birth:** 20/10/1997 **Nationality:** Pakistani

RESEARCH DIRECTIONS

Interests

- Artificial Intelligence and Machine Learning
- AI for Science (AI4S)
- Drug Discovery and Computational Biology
- Healthcare AI
- Large Language Models and AI Agents

PUBLICATIONS

[2026] **TransDTA: A Model for Drug-Protein Binding Affinity Prediction**

Reference: Hamza, M., Uddin, M. I., Aldhyani, T. H. H., Adnan, M., & Al-Adhaileh, M. (2026). TransDTA: A Model for Drug-Protein Binding Affinity Prediction.
First author manuscript completed; journal submission in progress

Authors: Muhammad Hamza, M. Irfan Uddin, Theyazn H.H. Aldhyani, Muhammad Adnan, Mosleh Al-Adhaileh

[2026] **Explainable Graph Neural Networks for Substructure-Level Toxicity Prediction on ClinTox**

Reference: Hamza, M., & Uddin, M. I. (2026). Explainable Graph Neural Networks for Substructure-Level Toxicity Prediction on ClinTox. International Conference on Cognitive and Generative Intelligence. (Submitted)
Authors: Muhammad Hamza, M. Irfan Uddin

[2024] **Federated Learning in Healthcare (Book Chapter)**

Reference: Hamza, M. (2024). Federated Learning in Healthcare. In M. I. Uddin & W. K. Mashwani (Eds.), Federated Learning: Unlocking the Power of Collaborative Intelligence. CRC Press / Taylor & Francis.
Authors: Muhammad Hamza | **Publisher:** Routledge Taylor & Francis Group

EDUCATION & TRAINING

[13/07/2023 - 21/05/2026] **Master of Science in Computer Science**

Kohat University of Science and Technology

City: Kohat | **Country:** Pakistan | **Final grade:** 3.10/4 CGPA | **Thesis:** TransDTA, a sequence-based deep learning model for drug–target binding affinity prediction, using a Transformer–BiLSTM drug encoder with pretrained-vocabulary tokenization, evaluated on the Davis and KIBA benchmarks

[01/10/2018 - 24/08/2022] **Bachelor of Science in Computer Science**

Government College University Faisalabad

City: Layyah | **Country:** Pakistan | **Final grade:** 3.29 CGPA | **Thesis:** Final-year project: full-stack cafe mobile app (React, Node.js, Firestore) with authentication and API protection.

WORK EXPERIENCE

Studio Fritz Gnad

Country: Germany

[01/10/2023 - Current] **Software Engineer**

- Build custom web applications and maintain containerized backend infrastructure for production use.
- Develop automated ML pipelines for data preprocessing, model integration, and inference.
- Integrate web3 services and fix performance bottlenecks in distributed systems.

SKILLS

AI & ML

Artificial Intelligence | Large Language Models | Machine Learning | Explainable AI | Graph Neural Network | Transformer Models | Deep Learning

Domain

Drug Discovery | Federated Learning | Bioinformatics

Programming

Python (computer programming) | Pytorch | NodeJS, | Git/Github, Docker, Gitlab

Engineering

Backend Development | Cloud Computing. | Containerisation (Docker & Kubernetes) | Distributed Systems

LANGUAGE SKILLS

Mother tongue(s): Saraiki | Urdu

Other language(s): English

RECOMMENDATIONS

Name: Dr. Muhammad Irfan Uddin Chairperson & Professor CS Department, University of Swabi, KPK

Strongly recommended based on performance

Email: irfanuddin@uoswabi.edu.pk

Name: Dr. Muhammad Adnan Senior Lecturer Institute of Computing, Kohat University of Science and Technology (KUST)

Strongly recommended based on performance

Email: adnan@kust.edu.pk

Name: Fritz Gnad Founder at Studio Fritz Gnad & Co-founder at Metonymy Machine gmbh

I believe that Muhammad will make a valuable addition to any organization that he may join. Strongly Recommended

Email: fritz@metonymy-machine.com